



DATA STRUCTURES AND ALGORITHMS

LECTURE 01-02

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WHAT IS DATA



The quantities, characters or symbols on which operations are performed by a computer , which may be stored or transmitted in the form of signals and recorded on any storage device .

FOR EXAMPLE

$$C = A + B$$

WHEN DATA BECOMES INFORMATION

DATA : an atmy iKe nam si

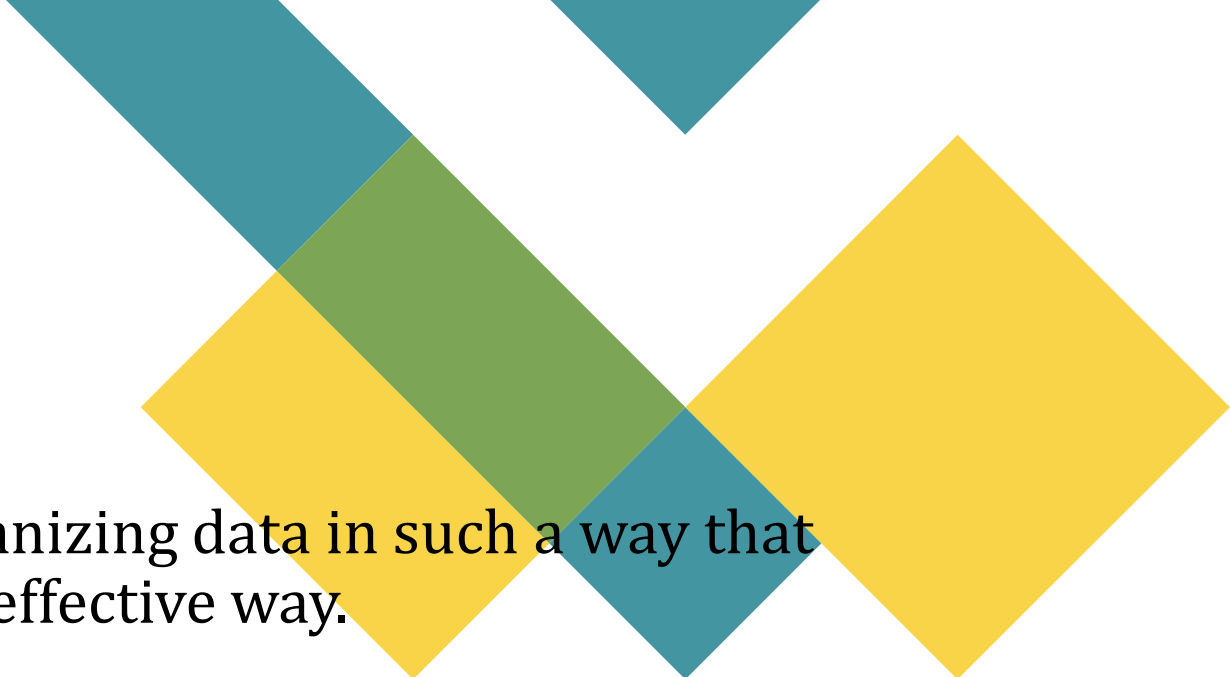
Collection of characters

INFORMATION : My Name Is Kainat

If data is arranged in a systematic way then it gets a structure and becomes meaning. Hence data needs to be managed in such a way that it gets us some meaningful information . To provide an appropriate way to structure the data.

We need to know about **DATA STRUCTURES**

DATA STRUCTURE



Data structure is a way of collecting and organizing data in such a way that we can perform operations on these data in effective way.

Data structure is about rendering data elements in terms of some relationship, for better organization and storage.

Data structures are structures programmed to store organized data, so that various operations can be performed in effective way.

It represents knowledge of data to be organized in memory.

CHARACTERISTICS OF DATA STRUCTURE

Correctness: Data structure implementation should implement its interface correctly

Time Complexity: Running time or execution time of operations of data structure must be as small as possible

Space Complexity: Memory usage of a data structure operation should be as little as possible.



CHARACTERISTICS OF DATA STRUCTURE

As applications are getting complex and data-rich, there are three common problems that application face now-a-days:

Data Search: Consider an inventory of 1 million items of a store. If the application is to search an item, it has to search an item in 1 million items every time, slowing down the search. As data grows, search will become slower.

Processor Speed: Processor speed although being very high, fall limited if data keep growing at exponential rate.

Multiple Requests: When thousands of users search an item simultaneously on a web server, even fast servers falls while searching the required data.

ALGORITHM



Algorithm is the representation of a solution to a problem.

It is step by step procedure to solve a given problem.

An algorithm contains finite number of instructions in order to solve a problem.

EXAMPLE



The following algorithm is given below.

Explanation:

Step 1: Place the fresh water in a saucepan or a kettle.

Step 2: Boil the water to match your style of tea.

Step 3: Put the black tea leaves in that pot

Step 4: After that add some milk into that pot.

Step 5: Add some sugar.

Step 6: Boil for some time.

Step 6: Ready to drink.

Thank you

